

applied to the block device.

The fourth field is the list of options that should be applied to the device when mounted to customise the behaviour. This field is required, and there is a set of commonly used options called defaults.

The last two fields are the dump flag and *fsck* order. The dump flag is used with the *dump* command to take a backup of the contents of the device. The *fsck* order field determines if the *fsck* should be run at the boot time, in the event that the file system was not unmounted cleanly. The value of the *fsck* order indicates the order in which file systems should have *fsck* run on them if multiple file systems are required to be checked.

Having an incorrect entry in */etc/fstab* may render the machine unbootable. To avoid that situation, an administrator should verify that an entry is valid by unmounting the file system back into place. If the *mount -a* command returns an error, it should be corrected before rebooting the machine.

Managing swap space

A swap space is an area of the disk that can be used with the Linux kernel's memory management sub-system. Swap systems are used to supplement RAM by holding inactive pages of memory. The combined RAM plus swap spaces is called virtual memory.

When the memory usage on a system exceeds a defined limit, the kernel will comb through the RAM looking for idle memory pages assigned to processes. The kernel writes the idle pages to the swap area, and will reassign the RAM page to be used by another process. If a program requires access to a page that has been written to disk, the kernel will locate another idle page of memory, write it to disk, and then recall the needed page from the swap area.

Since the swap areas reside on disk, swap is incredibly slow compared to RAM. When it is used to augment system RAM, usage of swap space should be kept to a minimum, whenever possible.

Creating a swap space

To create a swap space, an admin needs to do the following three things:

- Create a partition

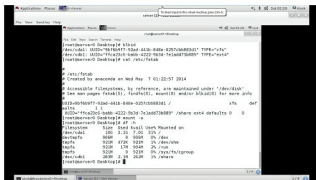


Figure 4: The entry of the *fstab* file for permanent mounting of a file using UUID

- Set the type of the partition as 82 Linux Swap
- Format a swap signature on the device

The creation of a partition and setting the type of the partition is shown in Figure 5.

Formatting the device

The *mkswap* command applies a swap signature to the device. Unlike other formatting utilities, *mkswap* writes a single block at the beginning of the device, leaving the rest of the device unformatted so that it can be used for storing memory pages.

```
# mkswap /dev/vdb2
```

Activating a swap space

An admin can use the *swapon* command to activate a formatted swap space. *swapon* can be called on a device, or *swapon -a* will activate all swap spaces available.

Logical volume management concepts

Logical volume and its effective management make it easier to handle disk space. If an LVM hosted file system needs space, it can be allocated to its logical volume from the free space in its volume group and the file system can be resized. If a disk starts to fail, a replacement disk can be registered as a physical volume with the volume group, and the logical volume extents can be migrated to the new disk.

LVM definitions

Physical storage devices preserve data in logical volume. These block devices can be either disk partitions or whole disks. They must be initialised as LVM devices in order to be used as an LVM.

Physical volume: These are used for underlying physical devices in volume groups. LVM automatically converts physical volumes into physical extents. Physical extents are the smallest chunks of data — the smallest storage block on a physical volume.

Volume group: This consists of one or more physical volumes. A particular physical volume can be allocated to only one volume group. A volume group can consist of unused space and it can have more than one LVM.

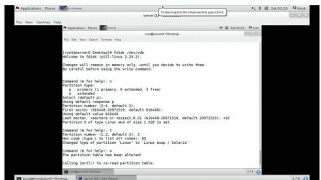


Figure 5: The creation of a swap partition